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BusDig - List. Probs

1) $A \cdot B + \bar{A}$

$$(\bar{A} + A)(\bar{A} + B)$$

$$1 \cdot \bar{A} + B$$

$$\bar{A} + B$$

Ⓒ

2) $A + (A \cdot \bar{B}) + (A \cdot B)$

$$A(1 + \bar{B} + B)$$

$$1 + B$$

$$1$$

A Ⓑ

3) $(A+B)(A+C)$

$$A + AC + AB + BC$$

$$A + AC$$

$$A + AB$$

$$A + BC$$

Ⓓ

4) $A \cdot (B + \bar{B}) \cdot C$

$$A \cdot 1 \cdot C$$

$$A \cdot C$$

Ⓐ

5) $(\bar{A} \cdot B) + (A \cdot B)$

$$B(\bar{A} + A + 1)$$

$$1$$

Ⓑ

6) $(A + A \cdot B) \cdot (B + A \cdot B)$

$$A(1 + B)$$

$$A$$

$$B(A + 1)$$

$$B$$

$$AB$$

Ⓒ

7) $(A \cdot B \cdot C) + (A \cdot B \cdot \bar{C}) + (A \cdot \bar{B} \cdot C)$

$$A(BC + B\bar{C} + \bar{B}C)$$

$$B(C + \bar{C})$$

$$B \cdot 1$$

$$B + \bar{B}C$$

$$A \cdot (B + C)$$

$$AB + AC$$

Ⓑ

8) $(A + \bar{B})$

$\bar{A} \cdot B$

(B)

9) $(\bar{A} \cdot B \cdot \bar{C})$

$\bar{A} + \bar{B} \cdot \bar{C}$

$A + \bar{B} + C$

(C)

10) $((A + B) \cdot (A + C))$

$(\bar{A} \cdot \bar{B}) + (\bar{A} \cdot \bar{C})$

$\bar{A}(\bar{B} + \bar{C})$

(A)

11) $((A \cdot B) + C)$

$\bar{A} + \bar{B} \cdot C$

(A)

12) $(\bar{A} + B + \bar{C})$

$A \cdot \bar{B} \cdot C$

(C)

13) $((\bar{A} + B))$

$\bar{A} \cdot \bar{B}$

(A)

14

A	B	C
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

$B + A$

15

A	B	C
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

$\bar{A} + BC$

(E)

16

A	B	C
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

BC

00	01	11	10
0	1	1	1
1	1	1	1

$AB\bar{C} + A\bar{B}C + \bar{A}B\bar{C}$

(C)

tilibra

17)

AB \ CD	00	01	11	10
00				1
01	1	1	1	1
11				1
10				1

 $\rightarrow \bar{A}B$ $AB + C\bar{D}$ $\rightarrow C\bar{D}$

(A)

18)

AB \ CD	00	01	11	10
00	1			1
01		1	1	
11		1	1	
10	1			1

 $\rightarrow \bar{D}\bar{D}$

(E)

 $\rightarrow BD$

19) $\bar{A} \cdot \bar{B} \cdot \bar{C} + \bar{A}B\bar{C} + A \cdot \bar{B} \cdot C + ABC$

0 0 0 0 1 0 1 0 1 1 1 1

A \ BC	00	01	11	10
0	1			1
1		1	1	

 $\rightarrow \bar{A}\bar{C}$

(B)

 $\rightarrow AC$

20)

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11			1	
10				

 $\rightarrow \bar{A} \quad \bar{A} + BCD$ $\rightarrow BCD$

(D)